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Website <https://u3aepsom.nz/>.

MEETING PLACE

Royal Oak Bowls, 146 Selwyn St, Onehunga

10am on the 2ND Thursday of most months

NEWSLETTER

AUGUST 2026

Next meeting

10-12noon

Thursday 13 August 2026

Late last month I attended the annual u3a Auckland Network event. This is a gathering together of all the u3a branches in Auckland. I'd never been before but everyone who had attended in the past urged me to go along because it was always such an enjoyable day. They weren't wrong. It was a good day. The day was structured around two main speakers: Dame Wendy Pye and the economist, Shamubeel Eaqub, plus four mini-speakers – all remarkably good (who knew what a “chopine” was?).

But, for me, the best part of the day was just chatting to people from other u3a groups. There are 28 in Auckland with memberships ranging from 65 to just over 300. There were 305 people at the event and, just like us, a wide diversity of backgrounds and interests. There was a great deal of curiosity about other groups. It seems that a lot of u3a members want to visit other groups when they meet to see how they do things. Not a bad idea. Maybe we should do that too.

I would encourage everyone to make a mental note and seriously consider going to next year's event. If it's as good as this year's, it'll be enjoyable.

Two final points.

If you do go next year, take your own coffee. Theirs was ghastly – a brew so insipidly vile they must have wrung out the cleaner's rag and strained some lukewarm water through it. And – I bet you just googled chopine.

Ian Jost
President

EPSOM U3A EXECUTIVE

President

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Immediate Past President:

Duncan MacDonald – 021 316 661

Secretary

Jenny Whatman – 027 353 2487
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Minutes Secretary

Jeanette Saunders – 624 5025

Membership Secretary

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General Duties

Kaye Buchanan- 021 299 0709

Almoner

Charmaine Strang – 027 4177 556

Interest Group Co-ordinators

Joslyn Squire – 021 168 0680

Bill Hagan – 021 611 247

Guest Speaker Organiser:

Ian Jost – 027 488 7037

Legal Advisor

Mike Matson – 022 630 7968

Newsletter

Jeanette Grant – 638 8566

Greeters:

Don Buchanan 0275 928 131

Helen Dallaston - 022-2647786

INTEREST GROUP CONVENERS

Applied Sciences

Peter Parsons - 021 521446

Attending Performing Arts

Shirin Caldwell – 630 1662

Architecture

Brian Murray – 021 026 68396

Art Appreciation

Kaye Buchanan – 021 299 0709

Art History

Emily Flynn – 021 0902 5094

Big History

Emily Flynn – 021 0902 5094

Book Chat

Helen Holdem - 021 260 3510

Chess Group

John Locke - 021-187 8061

Comparative Religions

Duncan MacDonald - 021-316 661

John Locke- 021-187 8061

Current Affairs

Shirley McConville – 622 3542

Fabric & Fibre Crafts

Charmaine Strang – 027-4177 556

Famous & Infamous Group

Shirley McConville – 622 3542

Foodies

Graham Gunn – 027 445 0929.

Garden Appreciation

New one needed

Introduction to Family History

Bryn Smith – 027 280 5235

Latin

Phyllis Downes – 630 5867

Lunch Bunch

Shirley McConville – 622 3542.

Music Appreciation

Carleen Edwards – 624 6298

History & Social Change

Helen Holdem – 021 260 3510

NZ History

Kaye Buchanan - 021 299 0709

Philosophy

Jocelyn Hewin - 634-1552

Scrabble

Joslyn Squire – 021 168 0680

Stories of the Past

Jeanette Grant – 638 8566

Te Reo Maori

Jenny Whatman – 027 353 2487

Travel

Diana Hart- 021 284 4402

JULY SPEAKER REPORT

Our July speaker, Colleen Brown, has had a demanding career as a teacher, Manukau City Councillor, and Counties Manukau DHB board member. She has also worked tirelessly to support South Auckland Youth and the disability sector, due to her concerns that they lack a voice in the media. For her work in the community and with disabled people, she received the KSO, but her great current passion is for writing.

Colleen contributes well-researched articles on social concerns to The Listener and her published work includes children's books, "Violet's Scarf" and "Elvie the Sheep", with a Book 2, "Elvie the Exceptional City Sheep" about to be available for purchase in August.

Colleen believes none of these books would have been as successful without the in-depth research she had to do for her adult book, "The Burford Kiwi". She says that "some stories announce themselves, while others whisper, waiting for someone to be curious enough to follow them". Her curiosity about NZ World War soldiers was prompted by a cake tin filled with memorabilia and a 1916 photo she found, of a relative called Bertie, who had served in Europe, completing a 250km march through Germany and occupied countries while suffering dreadful conditions. Rats ate their food, their clothing and boots were in tatters, their feet bled, there was poor shelter and inadequate food, no baths for six weeks and it was winter. Influenza and measles had also arrived.

Many of these suffering soldiers were sent to the south of England to recuperate after World War I, but Bertie died in 1919 while waiting for a ship to repatriate him back to NZ. There were many unanswered questions that led Colleen on a five-year search for answers. Colleen loves to talk to people and find their back-stories, so was eager to discover why the Kiwi men had to wait so long before being able to come home. Her search led her to use diaries, documents, old negatives and drawings, graveyards, visits to relatives and people with connections to the soldiers.

The soldiers were living in a camp built for 4500, but there was overcrowding with 6500 men being housed in the inadequate camp. She found there had been a riot in March 1919, causing damage to the tune of \$10,000 which had to be paid back to the Imperial Army, this leading to an inquiry and the dismissal of Brigadier-General Stewart. To fill in their time while waiting for a ship, the men got the idea of creating a massive Kiwi, carved into the chalk hillside near Burford, on Salisbury plain. The original Blenkarne drawing, possibly based on a kiwi in Buller's Book of NZ Birds, is now in the Waiouru Military Museum. The kiwi was laid out on Beacon Hill by Victor Low, an engineer (and also from the first Chinese family in NZ,) and it was completed in 1919. The kiwi had to be covered up while the war was still in its final stages, to stop the Luftwaffe from using it to locate the military camp.

By 1980 the kiwi had fallen into disrepair and The Prime Minister, Mr Muldoon, refused to spend money on repairing it, so Danny Fisher took on the job. This was no mean feat as the kiwi covered 1.25 acres, was 420 feet high, 150 feet long, and 35 feet wide, while the 249 letters on it to denote the squadron who carved it, are 20 metres long.

In 2019 a commemorative plaque was installed. Colleen thinks we should be proud of our history and should seek to record many of these events that may have faded into the past.
Joslyn Squire

MEMBERSHIP

Three new members have brought our current total up to 202.

MAIN SPEAKER

This month our speaker will be Ian Jost.

Ian gave a talk last September that focussed on the Medieval Battle of Agincourt in 1415 – an English victory that was considered at the time as being miraculous and a clear sign of God's favour for King Henry V of England.

This month, he will again visit the fifteenth century and investigates a pivotal moment in Henry's life; one often overlooked or trivialised by modern historians. It's a story that conflates rebellion, treason, courage, desperation, ingenuity, a look at medieval medicine that may surprise you, and perhaps another miracle.

INTEREST GROUPS

There is keen interest in a Literature group, which could meet on Friday mornings. A series of DVD's about writers will be shown, followed by discussion about members' reading. The Deaf Centre have agreed to be the venue.

2026 MEETING DATES.

13 August 10 September 8 October 12 November AGM

NB Always wear your name badge and be seated ready at 10am

SOLAR RANCH

Silicon Ranch, which owns a 40-acre farm in Christiana, outside of Nashville, believes cattle-grazing is the next frontier in so-called agrivoltaics, which mostly has involved growing crops or grazing sheep beneath the panels. The solar company debuted the project this year and will spend the next year working to demonstrate to farmers that much larger cattle also can thrive at solar sites. If successful, advocates say, that could jump-start new projects to meet the soaring electricity demand driven by rapidly expanding data centers — without contributing climate-warming carbon emissions — and help cattle producers hold onto their land and livelihoods.

"Solar is one of the most powerful tools we have for cutting emissions and ... is cost-competitive with fossil fuels," said Taylor Bacon, a doctoral student at Colorado State University. Solar panels often pivot to near-vertical angles to capture the sun's rays, leaving little room underneath for cattle; simply raising the panels is cost-prohibitive because of the amount of steel required. So Silicon Ranch raised the panels a little but also developed software that workers activate to turn the panels close to horizontal when cattle are grazing, giving them room to wander, said Nick de Vries, the company's chief technology officer.

Workers rotate the cattle — currently 10 cows and their calves — between paddocks every few days so panels on the ungrazed portion of the site operate normally, generating a supply of roughly 5 megawatts of electricity for Middle Tennessee Electric, a rural electric co-op. For solar companies, agricultural land is generally easier to develop than other types of sites. But many farmers — and communities — will need to be convinced that solar grazing will benefit them because of past practices that destroyed topsoil and took land out of production permanently.

"For many agricultural stakeholders, it is offensive to see high-quality farmland getting graded and piled when that's a farm family's legacy," said Ethan Winter, national smart solar director at American Farmland Trust. But he sees potential for solar grazing partnerships to help farmers keep their land in production and earn extra income.

WIND VILLAGE IN GERMANY

Feldheim's 130 residents draw their electricity and heating from a cluster of wind turbines on the village outskirts, a biogas plant fed by local corn silage and manure, a solar park on a former Soviet military site and a wood chip boiler as backup. Together, the installations produce way more energy than the village actually needs; the surplus current is sold to the national grid.

A ten-megawatt battery storage facility, part-funded by the European Regional Development Fund, keeps the local grid stable when conditions change.

Making it all work took some creative thinking — including building an entirely new electricity grid when the big utility companies refused to play ball. But for Michael Raschemann, the head of Energiequelle — the energy company behind the project — it proves that at this scale, energy self-sufficiency is not just possible — it's essential.

Feldheim's success was built on many factors: a good location, a small and close-knit community willing to think differently, visionary investors and support from national and European policy. It may not be a model that works everywhere. But as an example of what locally-owned renewable energy can achieve, it's hard to argue with the electricity bill.

LAKE REINCARNATION

For a brief stretch in California's San Joaquin Valley, roads, tomato fields, cotton land and pistachio orchards gave way to blue water. The San Joaquin Valley was a dry agricultural region where the old lakebed had long been converted into farmland.

That water was Tulare Lake, also known as Pa'ashi to the Tachi Yokut Tribe. Tulare Lake once stretched more than 100 miles long and more than 30 miles wide, - the largest body of fresh water west of the Mississippi River and one of the largest freshwater bodies in the western United States. The lake was not mainly fed by local rainfall but was supplied primarily by snowmelt from the Sierra Nevada mountains. Because the valley has no natural outlet, water can collect in the basin when enough snowmelt and runoff arrive. The lake had been drained into farmland more than a century earlier. Then, in February 2023 the lake returned after a series of severe weather events, including atmospheric rivers that brought heavy rain and snowfall.

By spring 2023, the lake covered more than 100,000 acres, according to The Guardian's March 2024 report. Local station FOX26 News reported that Tulare Lake peaked at roughly 120,000 acres, covering more than 10% of Kings County.

Less than a year later, much of it was gone again.

THUMBNAIL-SIZED THRUSTERS COULD TAKE CUBESATS TO MARS

Engineers at MIT have successfully tested a hybrid propulsion system for small satellites that combines the raw punch of chemical rockets with the long-haul efficiency of electric thrusters – and crucially, both run on the same fuel. This tech could propel cheap, briefcase-sized spacecraft to Mars and beyond.

Microsatellites – roughly the size of a shoebox or carry-on bag – are cheaper and easier to launch than conventional satellites, but they've always faced a frustrating constraint. If you wanted one to perform quick orbital maneuvers and slow, precise adjustments, you needed two separate propulsion systems with two separate fuel sources. In a platform so small, that presents quite a challenge.

Chemical thrusters burn fuel fast and generate a powerful, rapid thrust that is key for changing orbit in seconds. Electric thrusters (specifically electrospray thrusters) are extraordinarily fuel-efficient, but slow. They can be used for making precise adjustments, or to accelerate a spacecraft gradually over months or years, making them ideal for long interplanetary voyages.

These technologies are complementary but have remained incompatible in a small format – at least until now. The breakthrough came from the US Air Force, which had spent years developing a propellant called ASCENT (Advanced SpaceCraft Energetic Non-Toxic propellant) as a greener replacement for hydrazine, a highly toxic chemical traditionally used in spacecraft propulsion. ASCENT was designed for chemical thrusters, but it had a hidden property nobody had yet exploited: it's an ionic liquid – essentially a salt that stays in liquid form even in the vacuum of space which makes them perfect for electrospray thrusters, which work by applying an electric field to a liquid, charging its ions, and firing them out as a fine jet that generates thrust.

In testing, ASCENT delivered a thrust-to-power ratio of 40–65 micronewtons per watt, a specific impulse (a standard measure of propellant efficiency) of 600 seconds, and an overall efficiency of 15%. The thrusters ran continuously for up to 167 hours with no measurable degradation. The next step is orbit. MIT is working with NASA on the Green Propulsion Dual Mode mission – a CubeSat carrying one chemical thruster and four electrospray thrusters, all fed by a single shared ASCENT tank. A launch is planned for later this year.

If it works, the applications range from the practical – deploying satellite constellations to track a hurricane in real time – to the audacious. "We could send CubeSats to Mars, or the asteroid belt, where they could make the journey slowly using electrospray thrusters," Lozano adds. "You could then use your chemical thrusters to quickly move to look at interesting features."

Source: MIT

FINDING FIRE

Just when our ancestors fanned their first flames is an ongoing question, with current evidence suggesting it had to be at least a million years ago. A discovery reported in the journal PLOS One pushes this timeline back even further, confirming hominins were probably playing with fire far earlier.

By testing the luminescence of tiny bones in ancient owl pellets, an international team of researchers led by Dores Marin-Monfort, a paleontologist from the National Museum of Natural Sciences in Spain, has provided strong evidence that residents of South Africa's Wonderwerk Cave may have been warming their toes in front of a raging fire some time between 1.07 and 1.78 million years ago.

Wonderwerk Cave is already a renowned site among paleontologists, thanks to its extensive use by ancestral humans. For decades, researchers have sifted through almost 20 feet (about six meters) of material containing everything from pollen to stone artifacts to burned bones.

It's those charred remains – along with residual ash and heat-altered stone – that have previously set records for the world's oldest signs of a campfire halfway through the Pleistocene. Evidence of burned material isn't necessarily evidence that the burning is intentional. Translocation of charred sediments can also mess up timelines, making it even more difficult to nail down when humans mastered the art of a good roast. That makes the heat-affected remains of fire pits and cooked animals gold standards for confirming signs of a tamed flame.

When it comes to the oldest examples of fire use, however, hearths are extremely hard to find, implying that the first fires were opportunistic rather than deliberate constructions. While it's possible to use a bone's color to tell if it has been cooked, metal contaminants in the soil can cause similar changes in hue. A suite of spectrographic tests is commonly used instead, including Fourier Transform Infrared Spectroscopy. Yet these also have their limitations in capturing the extent of a bone's charring.

Marin-Monfort and her team also applied a non-invasive method of measuring the luminescence of around 160 small mammalian bones found in owl pellets, which once littered the cave floor "like a carpet".

Their findings confirm that small fires were located deep within Wonderwerk Cave hundreds of thousands of years before previously established records, in line with other contested studies that also suggest earlier dates.

For cooking, warmth, or light, nobody can be sure. But the fires were unlikely to have been the result of a stray spark from a passing wildfire, the researchers write, leaving the strong possibility that these were deliberately placed by an ancestor such as *Homo erectus*. It would take hundreds of thousands more years before fire would become a commonly used tool among our own species. But our fascination with fire, it seems, has been flickering away in our mind since the very beginning.



U3A Meadowbank Winter Series 2026

Book now, and bring a friend, or two.

New Zealand – Past, Present and Future

When: Three successive Friday mornings in August, the 7th, 14th and 21st, 10 am.
Place: St Chads Church, 38 St Johns Road, Meadowbank
Presenters: Peter Lineham, Stephen Child and Brian Easton
Topic: New Zealand: Past, Present and Future
Cost: \$60.00 for three sessions
Including: Morning tea

Dates, Speakers and Topics :

7th August 2026, Prof Peter Lineham MNZM - Education

14th August 2026, Dr Stephen Child - Health

21st August 2026, Brian Easton NZAE – Economics

Bookings:

Book by paying \$60.00 for each attendee to the U3A Meadowbank bank account Kiwibank account number 38-9014-0359752-00. Enter “winter series” and your name as reference. Please email u3ameadpres@gmail.com for confirmation of booking.